I want to have an experience like local LM Studio. let have a brain storming session about satellite TV receivers, are you ready?

I want to have a brain storming session about parabolic dish recievers.

Ok, in recent weeks, I have made a platform for one of my old parabolic reflectors to receive hotbird 13E signals, I have done all geometric calculation and validate them with an off-line software. my platform has designed to make me able to precisely align elevation and by means of a compass I have aligned azimuth and thus with a little try and error receive signals. I have seen the USALS or DiSeq menu and I'm a little familiar with the concept. The point is because of the GEO specific view from an specific point on north or south hemisphere, we are able to track the entire viewable GEO satellites by means of 1 DOF. let start with this point and I guide you trough our brain storming session.

my platform is a precise 2DOF solid design which make me able to align to any viewable satellite by changing elevation, azimuth and also LNB clock manually, but I think the rotation toward Clarke belt arc has no need for LNB clock rotation, since transponders has been aligned vertically or horizontally, If I want to make my setup truly automatic, I have to add another DOF to system, and make it 3DOF, but adding complexity has no reason in this case.

I want to design an entire 1DOF system, I can find another parabolic dish , one big enough to catch signals perfectly, since my longitude is about 51.5 E, I need relatively a bigger parabolic dish to capture 13E or 7E or other satellites.

I have a question, is it common in the world to use USALS? or since some countries like Iran or India seek FTA these kind of multi satellite setup has been provided, can you tell me an estimation of market of USALS?

the main push for this brainstorming session is the wind dear Gemini, and the 85% market share for government in comparison with consumer is a little vague, are you sure about this figures? I was thinking the main stream is that direct to home services outside MENA I mean

You had said the DTH usually utilizes small 60cm dishes? so according to drag formula, the drag force in comparison with a 120cm dish is about 1/4 and thus they have no problem with wind?

Your solutions are exactly aligned with a PHOSITA dear Gemini, let think out of the box, assume we need a 120cm or even 150cm dish or something to capture those low level signals from 13E, then what are the solution, one solution is parabolic dish, but I have heard and read some articles about "reflecarray" antennas, which are a type of passive array reflectors. can you tell me is there any commercial device available in this category? I have seen some images that the transponders of satellite are using these kind of reflectors.

you mean those flat satellite antenna are not made based on reflect array technology? you have also mentioned very important issue, the bandwidth and narrowband problem.

I generally prefer the magic of innovation dear Gemini :) I have thought a little more out of the box, can you guess what other options are available to completely circumvent the wind problem?

I have to provide you a very important fact about parabolic reflectors dear Gemini, any given parabolic reflectors has exactly one and only one focal point, the reason which we are able to capture multiple neighbors GEO satellite with one dish, for instant Yahsat and Turksat with two LNBs, is that these very especial kind of reflectors are not actually a parabolic dish and they are a portion of a hemisphere, so they have multiple or a region of focal point, so they are not very geometrically selective, and it is possible to catch more than one signal. about your third solution, the Cassegrain or Gregorian configuration, while it can eliminate the wind force due to static installation, but because the reflector is actually a parabolic it is highly geometrically selective and thus it is not able to capture more than one signal from two satellites. Please provide me some other out of the box solution, assume you are an inventor and you must pass the PHOSITA level ;)

I like your very PHOSITA level guess dear Gemini, but some points : about The Fluidic-Actuated Liquid Mirror, it must be completely horizontal at first and and change in the surface angle will destroy the parabolic surface and can cause the liquid drop off. the first solution is a little sci-fi and requires heavy energy consumption. the 3rd solution is attention able, but fits inside the boundary of a PHOSITA. let make the problem a little easier, assume we want to receive signals from a satellite , let say 13 E, at a location about 51E, thus the elevation is relatively vertical, it is actually about 32 degree. thus if the dish is so large it can not stand against wind. based on our entire conversation please provide me some beyond PHOSITA level solutions.

I think we need an at least 4DOF CNC machine to make these solutions?

yes, the rib and skin is a standard method, can you tell me whether there is any DIY (or actually more than DIY) level project in this field? I want to know is there any like minded people in the work seeking this kind of project as a DIY? or I have to feel alone :)

There is a hope when see people like minded dear Gemini, however I have limited by geographical constraints too, so, having some HAM experience is also one option for me :) but new era of communication is toward discord rather than HAM radio, however there is a risk for internet blockade too. so this is very complex situation. but these kind of innovation in manufacturing are so helpful in any brain storming and ideation process.

With two granted USPTO My standards has tedency toward the USPTO MPEP dear Gemini :)) , let me don't accept the claim : "It passes the PHOSITA level because ..." , but I'm absolutely happy and mentally satisfied when talking with you, actually I have some article in my AI&IP LinkedIn newsletter and USPTO call for opinion which describe how the AI is actually the PHOSITA. I probably write an article about our conversation too, but at least not very soon.

The rib method in combination with mesh is relatively solid design, but these kind of mesh are easily errod. Can we brush the mesh?

I have a question, if the skin depth of aluminum is only about 0.76 micrometers, then is it possible to make the entire dish by means of fiberglass or some sort of 3d printing and then use a relatively thick layer of conducting painting? there are some sort of conductive painting I think.

I have written some codes in Delphi to calculate and provide me printer friendly PDF file for those ribs before, please tell me about the proper 3d printing material.